Baseline V2 Training and Evaluation Report

Adaptive Indian Road Sign Recognition Using Open-Set and Few-Shot Learning

Measured closed-set Baseline V2 run. No post-test tuning or retraining was performed.

Item	Value
Run ID	20260901_000220_558663_v2_mobilenetv3_small_100
Python	3.11.16
PyTorch	2.5.1+cpu
timm	1.0.29
Device	cpu
Deterministic	True
Train / validation / test	287 / 62 / 63
Classes	17
Best epoch	21
Best validation macro F1	0.622307
Test accuracy	0.603175
Test macro F1	0.675581

Pre-training verification and class weights

Review: 530 rows; 412 approved; 118 rejected; zero pending. Leakage checks passed for review IDs, source images, and perceptual groups. Excluded classes one_way, stop, and y_junction were absent. Locked file hashes are recorded in pretraining_verification.json.

Weight formula: N / (C * n_c); source: train split only.

Class	Train count	Weight
filling_station	15	1.125490
gap_in_median	9	1.875817
give_way	12	1.406863
hairpin_bend_ahead	16	1.055147
left_curve_ahead	18	0.937908
major_road_ahead	11	1.534759
maximum_speed_limit_30_km_h	11	1.534759
maximum_speed_limit_40_km_h	14	1.205882
maximum_speed_limit_80_km_h	13	1.298643
no_entry	10	1.688235
no_right_turn	8	2.110294
pass_either_side	8	2.110294
pedestrian_crossing_ahead	44	0.383690
right_curve_ahead	22	0.767380
road_hump	37	0.456280
school_ahead	26	0.649321
side_road_right	13	1.298643

30-epoch training history

Epoch	Train loss	Train acc.	Val. loss	Val. acc.	Val. macro F1	LR	Seconds
1	2.8635	0.0488	2.8916	0.0645	0.0216	0.00049863	23.9
2	2.8039	0.3589	2.8502	0.1613	0.1906	0.00049455	18.2
3	2.7357	0.5087	2.8198	0.2581	0.2500	0.00048779	18.4
4	2.6673	0.6829	2.7825	0.2581	0.2941	0.00047843	19.2
5	2.5880	0.7387	2.7216	0.3065	0.3529	0.00046657	21.6
6	2.5081	0.8049	2.6859	0.3871	0.4514	0.00045235	23.5
7	2.4270	0.8780	2.6184	0.3871	0.4315	0.00043591	24.8
8	2.3437	0.9268	2.5764	0.4677	0.5355	0.00041745	21.9
9	2.2649	0.9582	2.5565	0.4355	0.4825	0.00039715	21.9
10	2.1923	0.9686	2.5210	0.4839	0.4366	0.00037525	22.1
11	2.1205	0.9861	2.4899	0.4677	0.5031	0.00035198	25.8
12	2.0540	0.9930	2.4225	0.5484	0.5741	0.00032760	31.3
13	1.9984	0.9930	2.3932	0.5161	0.5066	0.00030237	30.6
14	1.9453	0.9895	2.3705	0.5000	0.5468	0.00027658	23.0
15	1.8775	1.0000	2.3537	0.5484	0.5535	0.00025050	18.1
16	1.8272	1.0000	2.3477	0.5000	0.5499	0.00022442	16.9
17	1.7908	0.9965	2.3354	0.5323	0.5917	0.00019863	17.0
18	1.7490	1.0000	2.2542	0.5484	0.5819	0.00017340	16.6
19	1.7166	1.0000	2.2303	0.5484	0.5999	0.00014902	16.3
20	1.6927	0.9965	2.2305	0.5323	0.6054	0.00012575	16.7
21	1.6663	1.0000	2.2317	0.5484	0.6223	0.00010385	17.1
22	1.6457	1.0000	2.2261	0.5161	0.5635	0.00008355	16.3
23	1.6317	1.0000	2.2069	0.5323	0.5777	0.00006509	16.1
24	1.6138	1.0000	2.1945	0.5323	0.6112	0.00004865	16.5
25	1.6117	1.0000	2.1824	0.5484	0.6151	0.00003443	32.3
26	1.6022	1.0000	2.1916	0.5484	0.6034	0.00002257	28.3

Epoch	Train loss	Train acc.	Val. loss	Val. acc.	Val. macro F1	LR	Seconds
27	1.6014	1.0000	2.1939	0.5484	0.6153	0.00001321	28.7
28	1.5930	1.0000	2.1919	0.5484	0.6153	0.00000645	22.6
29	1.5893	1.0000	2.1878	0.5484	0.6151	0.00000237	25.2
30	1.5906	1.0000	2.1878	0.5484	0.6073	0.00000100	22.2

Locked test results

Metric	Value
Top-1 accuracy	0.603175
Macro precision	0.735273
Macro recall	0.672549
Macro F1	0.675581
Weighted precision	0.617478
Weighted recall	0.603175
Weighted F1	0.586111
Correct / total	38 / 63

Class	Precision	Recall	F1	Correct/total
filling_station	1.000	1.000	1.000	3/3
gap_in_median	1.000	0.500	0.667	1/2
give_way	1.000	0.333	0.500	1/3
hairpin_bend_ahead	0.571	1.000	0.727	4/4
left_curve_ahead	0.000	0.000	0.000	0/4
major_road_ahead	1.000	0.667	0.800	2/3
maximum_speed_limit_30_km_h	0.667	0.667	0.667	2/3
maximum_speed_limit_40_km_h	1.000	0.667	0.800	2/3
maximum_speed_limit_80_km_h	0.600	1.000	0.750	3/3
no_entry	1.000	1.000	1.000	2/2
no_right_turn	1.000	1.000	1.000	1/1
pass_either_side	1.000	1.000	1.000	1/1
pedestrian_crossing_ahead	0.462	0.600	0.522	6/10
right_curve_ahead	0.400	0.500	0.444	2/4
road_hump	0.400	0.500	0.444	4/8
school_ahead	0.400	0.333	0.364	2/6
side_road_right	1.000	0.667	0.800	2/3

Confusions, embeddings, and limitations

Most common non-diagonal confusion pairs:

True class	Predicted class	Count
give_way	right_curve_ahead	2
pedestrian_crossing_ahead	road_hump	2
pedestrian_crossing_ahead	school_ahead	2
road_hump	pedestrian_crossing_ahead	2
school_ahead	hairpin_bend_ahead	2
school_ahead	road_hump	2
gap_in_median	pedestrian_crossing_ahead	1
left_curve_ahead	hairpin_bend_ahead	1
left_curve_ahead	maximum_speed_limit_30_km_h	1
left_curve_ahead	pedestrian_crossing_ahead	1

Embedding diagnostic	Value
Dimension	1024
Mean within-class cosine	0.558831
Mean between-class cosine	0.105743
Nearest-centroid closed-set accuracy	0.777778

Warnings and limitations

no_right_turn and pass_either_side each have only one test image, so their class metrics are unstable. Several other class supports are also small. The nearest-centroid value is a resubstitution diagnostic because centroids use the same test embeddings, including each query; it is not an unbiased model score. Softmax confidence is closed-set confidence, not an unknown-sign score. No unknown/OOD threshold was implemented. V1 and V2 are descriptive, not directly comparable benchmarks.